TEXTURE MAPPING AND GRAPHICS PIPELINE 4

CS 184: FOUNDATIONS OF COMPUTER GRAPHICS

1 A Texture and Mipmap Adventure

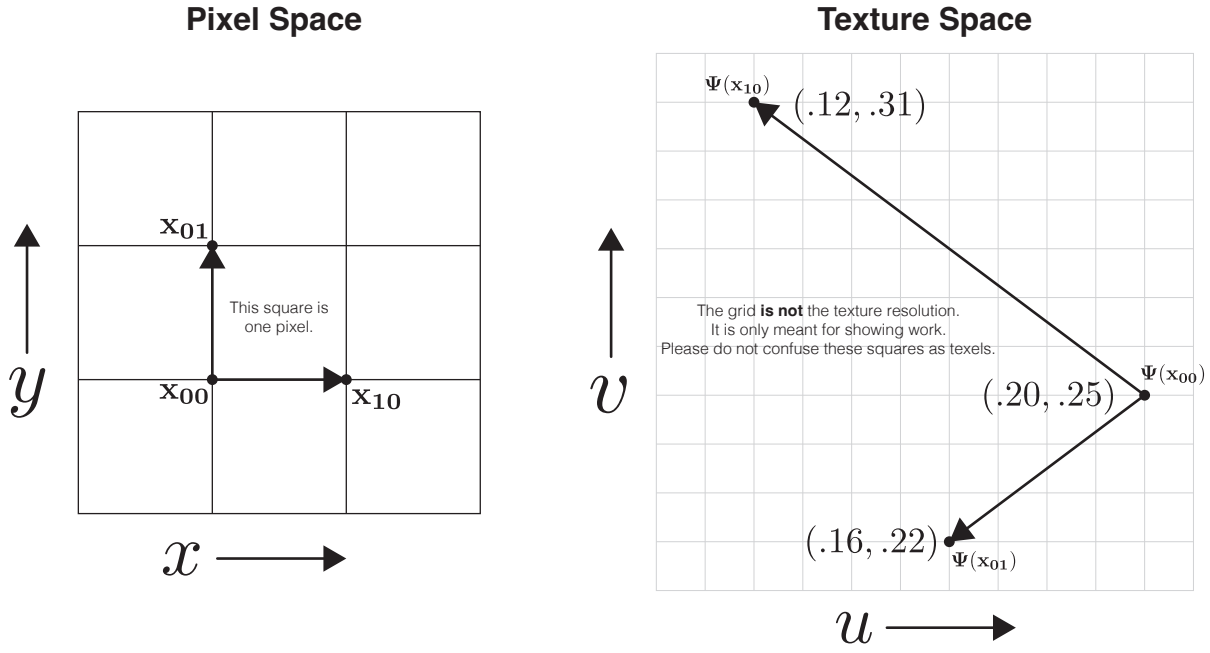
In this problem, we explore texture coordinates and mipmaps. A pixel may cover more or less than one texel, leading to aliasing. To handle this, we use mipmaps—pre-filtered, downscaled versions of the texture.

Consider an image mapped to a texture with a resolution of $(128, 128)$. We'll work through sampling a pixel and retrieving its value from the appropriate mipmap level. Note that the texture coordinates (u, v) lie in the range $[0, 1]$, not the texture resolution range.

1. **Conceptual Question:** What is the resolution of the mipmap at level 3?

2. **Conceptual Question:** Derive an expression for the amount of space needed to store all mipmap levels, relative to the memory needed to store the original texture.

3. We are given pixel point $\mathbf{x_{00}} = (x, y)$, as well as $\mathbf{x_{01}}$ and $\mathbf{x_{10}}$, each 1-pixel away from $\mathbf{x_{00}}$. These three points are mapped to their corresponding locations in texture space by the mapping $\mathbf{\Psi}(\mathbf{x})$. *Note that the mapping uses barycentric coordinates!* The texel space is in the range $[0, 1]$. At what mipmap level, L, should we sample to retrieve the texture for point $\mathbf{x_{00}}$?



4. For mipmap levels 0 through 5, the texture value at $\mathbf{\Psi}(x_{00})$ is given by:

$$T_0 = 0.38, \quad T_1 = 0.42, \quad T_2 = 0.36, \quad T_3 = 0.40, \quad T_4 = 0.39, \quad T_5 = 0.37.$$

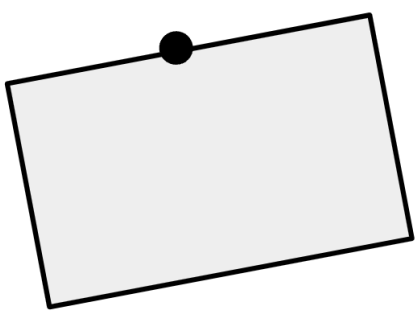
   Given our answer to the previous question, what texture value should we use?

5. How can we combine values from two neighboring mipmap levels to get an even smoother result?

2 Graphics Pipeline — Lightning Round!

1. Name and describe the three terms in the Blinn-Phong Reflection Model.

2. A light source shines on a tilted surface. Draw the light direction vector, **l**, and the normal vector, **n**, at the given point.



3. What is the light per unit area on this surface proportional to, according to Lambert's cosine law?

4. Complete the following implementation of the Z-Buffer Algorithm in C++.

```
const int WIDTH = 800; // Width of framebuffer
const int HEIGHT = 600; // Height of framebuffer

struct Color {
    float r, g, b;
};

struct Sample {
    int x, y;
```

```
    float z;
    Color color;
};

struct Triangle {
    std::vector<Sample> samples;
};

void zBufferAlgorithm(const ____________________________________ triangles,
                      ____________________________________ framebuffer,
                      ____________________________________ zbuffer) {
    for (const Triangle& T : triangles) {
        for (const Sample& sample : T.samples) {
            int x = sample.x;
            int y = sample.y;
            float z = sample.z;

            if (x >= 0 && x < WIDTH && y >= 0 && y < HEIGHT) {
                if (_____________________________________) {
                    framebuffer[x][y] = sample.color;
                    _____________________________________;
                }
            }
        }
    }
}
```

5. Prior to running this algorithm, what should the Z-buffer values be initialized to?